

Vadivelan Murugesan

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Summary

Robotics engineer with experience in ROS 2, C++, and Python across perception, SLAM, motion planning, and control. Builds and debugs real robots end to end, from drivers and embedded boards to planning and vision pipelines.

Skills

Programming:	Python, C, C++, Lua
Robotics & Control:	ROS 2, MoveIt, Control Systems, Kinematics, Teleoperation
Embedded Systems:	STM32, Arduino, Raspberry Pi, Jetson Orin Nano, ODrive, Serial Protocols (CAN, SPI, I2C), PCB Debugging
Perception & CV:	OpenCV, ArUco, YOLOv8, ICP, RANSAC
ML & Frameworks:	PyTorch, TensorFlow, Scikit-learn, CUDA, Autoencoder
Simulation Tools:	Gazebo, CoppeliaSim, MATLAB, Simulink
Sensors & Actuators:	RGB and Depth Cameras, IMU, LiDAR, Actuators (Servo, BLDC)

Education

New York University , MS in Mechatronics and Robotics	Sep 2023 - May 2025
Anna University, Chennai , BE in Electronics and Communication	Sep 2018 - June 2022

Experience

Research Scholar, ASAS Lab, NYU – New York City, NY Aug 2025 - Present

- Implementing a mini Autoware stack on Jetson Orin Nano for a unicycle model mobile robot, building the stack to be plug and play across other platforms with minimal changes.
- Integrating IMU, LiDAR, depth camera, wheel odometry, and motor control in ROS 2 with standardized topics, transforms, and timing.

Robotics Engineer, Surinova Pvt Ltd – Chennai, India Nov 2022 - Aug 2023

- Integrated STM32F4 with ODrive over CAN, SPI, and I2C to drive multiple BLDC motors with real time feedback and control.
- Built a hardware in the loop testbed with multiple ODrives and an STM32 to validate firmware and synchronize actuators before field tests.
- Used a PicoScope and debug scripts to isolate encoder and SPI noise, cutting troubleshooting time by about 50%.

Technical Intern, Cologenes Healthcare Pvt Ltd – Salem, India July 2021 - Sep 2021

- Automated a collagen sheet production step with a laser process, reducing processing time by 30% and boosting throughput by 20%.
- Implemented computer vision for alignment, improving accuracy by 80%.

Research Project Intern, e-Yantra Lab, IIT Bombay – Mumbai, India May 2021 - July 2021

- Collaborated with the project lead and team to explore simulation environments, develop solutions for real-world problems, and document results.
- Explored plugins and middleware including BlueZero and CoppeliaSim to assess feasibility for hardware implementation.

Projects

Proactive Human Manipulator Collision Avoidance GitHub

- Built a digital twin in Gazebo with real time 3D human pose estimation using an RGB D camera to simulate and test safe manipulator behavior in dynamic, human populated settings.
- Integrated ROS 2, MoveIt, and a UR16e robotic arm for adaptive, human aware motion control.

- Implemented an A RRT* planner for dynamic path planning, enabling safe trajectory replanning based on predicted human motion.

Automated Object Retrieval Mobile Robot

GitHub

- Developed a mobile robot with camera and Raspberry Pi for real time object identification using OpenCV; designed detection and localization.
- Engineered and programmed a two DOF arm to autonomously navigate and relocate objects within an arena, demonstrating control and planning in dynamic environments.

Color Cube Sorting Robotic Arm

GitHub

- Built a four DOF robotic arm with a gripper to sort color cubes on a conveyor, integrating a color sensor for real time identification and an ultrasonic sensor for proximity.
- Improved efficiency by utilizing cogs on the Parallax Propeller Board for multiprocessing, enabling simultaneous color and distance sensing.

Teleoperation of a 4 DOF Robotic Arm

GitHub

- Designed and prototyped a 4 DOF robotic arm controllable via a puppet rig with four potentiometers, similar to industrial pick and place.
- Added program memory for up to ten reprogrammable positions with simple save and cycle for flexible operation.
- Built a real time mirror mode that replicates the puppet arm's orientation for live teleoperation.

Key Projects in Perception, Machine Learning, and 3D Mapping

GitHub

- Developed custom RANSAC plane fitting and Iterative Closest Point for 3D point clouds to achieve accurate plane extraction and point cloud registration.
- Built augmented reality visualizations in OpenCV with ArUco markers, projecting 3D objects onto real world tags with calibrated rendering.
- Implemented object detection and tracking with YOLOv8 and a tracking pipeline to maintain persistent IDs across frames in dynamic video.

Robot Perception: Solving a 3D Maze Using Graph Networks

- Built a Spatial Vicinity Graph that treats each point in the 3D maze as a node with edges formed from global image descriptors, enabling shortest path planning with 100 percent accuracy in tests.
- Used SuperPoint and SuperGlue for feature extraction and matching with geometric verification to identify the destination node for precise navigation.
- Evaluated COLMAP, pose graph optimization with GTSAM, and Stella VSLAM before converging on a vision plus graph theory approach that balanced accuracy and runtime.

Leadership & Awards

Robotics Club of CEG, Chennai, India

2020 – 2022

President (2021–2022); Junior Office Bearer (2020 – 2021)

- Led 50+ members, established a structured learning track, and organized workshops and webinars.
- Initiated a multi-year delivery drone project and oversaw development of a gesture-controlled robot.

The Model School, Abu Dhabi, UAE

2017 – 2018

Student Council President

- Chaired student council meetings and represented students at interschool events.
- Founded new student clubs to expand co-curricular engagement.

Awards

- **eYantra Robotics Competition** — 3rd Place, 2021
- **Line Tracer Bot Competition** — 1st Place, 2020
- **Most Innovative Project Award** — 2017